

Name: \_\_\_\_\_

Date: \_\_\_\_\_

# PROPORTION, (DIRECT PROPORTION)

**LO: I can solve problems involving direct proportion using the unitary method and formulae.**

## Direct proportion

Two quantities are in direct proportion if when one doubles, the other doubles too. Their ratio stays constant.

Unitary method: Find the value of ONE unit first, then multiply to find the amount you need.

## Quick examples

|   |                                                    |                                |
|---|----------------------------------------------------|--------------------------------|
| ● | If 3 pens cost £6, then 1 pen costs £2             | unitary method                 |
| ● | $y = kx$ means $y$ is directly proportional to $x$ | formula                        |
| ● | Doubling $x$ doubles $y$ in direct proportion      | constant ratio                 |
| ● | If 3 pens cost £6, then 5 pens cost £8             | added 2 instead of using ratio |

## Key idea

If  $y$  is directly proportional to  $x$ , then  $y = kx$  where  $k$  is the constant of proportionality.

$$y = k \times x$$

$$k = y \div x \text{ (the constant ratio)}$$

## MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

### Example 1 - unitary method

5 books cost £35. Find the cost of 8 books.

$$\text{Cost of 1 book: } 35 \div 5 = \text{£}7$$

$$\begin{aligned} \text{Cost of 8 books: } &7 \times 8 \\ &= \text{£}56 \end{aligned}$$

$$= \text{£}56$$

Find the cost of one first, then multiply.

### Example 2 - finding $k$

$y$  is directly proportional to  $x$ . When  $x = 4$ ,  $y = 20$ . Find  $y$  when  $x = 7$ .

$$y = kx, \text{ so } k = 20 \div 4 = 5$$

$$y = 5x$$

$$\text{When } x = 7: y = 5 \times 7 = 35$$

$$y = 35$$

Find  $k$  first, write the formula, then substitute.

### Example 3 - scaling

A recipe for 4 people uses 300g flour. How much for 6 people?

$$\text{For 1 person: } 300 \div 4 = 75\text{g}$$

$$\begin{aligned} \text{For 6 people: } &75 \times 6 \\ &= 450\text{g} \end{aligned}$$

$$= 450\text{g}$$

Scale down to 1, then up to the number needed.

### Example 4 - direct proportion squared

$y$  is directly proportional to  $x^2$ . When  $x = 3$ ,  $y = 36$ . Find  $y$  when  $x = 5$ .

$$y = kx^2, \text{ so } 36 = k \times 9, k = 4$$

$$y = 4x^2$$

$$\text{When } x = 5: y = 4 \times 25 = 100$$

$$y = 100$$

Proportional to  $x$  squared means  $y = kx^2$ .

## YOUR TURN – PRACTICE (deliberate practice)

### 1. Unitary method

Find the value of one, then scale up.

- a) 3 pens cost £4.50. Cost of 7?
- b) 5 kg costs £12.50. Cost of 8 kg?
- c) 4 tickets cost £60. Cost of 10?
- d) 6 oranges cost £1.80. Cost of 9?

### 2. Finding the constant

y is directly proportional to x. Find k and the missing value.

- a)  $x = 3, y = 12$ . Find y when  $x = 5$
- b)  $x = 4, y = 28$ . Find y when  $x = 6$
- c)  $x = 10, y = 45$ . Find x when  $y = 27$
- d)  $x = 2, y = 9$ . Find y when  $x = 8$

### 3. Scaling recipes

Scale each ingredient.

- a) Recipe for 4: 200g flour. Amount for 10?
- b) Recipe for 6: 450ml milk. Amount for 4?
- c) Recipe for 8: 320g sugar. Amount for 5?
- d) Recipe for 3: 150g butter. Amount for 12?

### 4. Direct proportion problems

Solve each proportion problem.

- a) A car uses 6 litres for 72 km. How far on 10 litres?
- b) 3 workers build a wall in 12 hours. How long for 1 worker?
- c) £1 = 1.15 euros. Convert £45 to euros.
- d) A map scale: 2 cm = 5 km. What is 7 cm in km?

### 5. Mixed direct proportion

Solve each problem.

## COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) Direct proportion means  $y = kx$  ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

b) In direct proportion, as x increases y decreases ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

c) The graph of direct proportion passes through the origin ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

d) If 3 costs £6, then 5 costs £6 + £2 = £8 ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

e)  $k = y \div x$  is the constant of proportionality ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

## 6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) The cost of fuel is directly proportional to the volume bought. **25 litres** costs **£36.25**. Find the cost of **40 litres**. If you have **£50**, how many litres can you buy (to 1 d.p.)?

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

b) y is directly proportional to  $x^2$ . When  $x = 4, y = 80$ . Find: (i) the formula, (ii) y when  $x = 6$ , (iii) x when  $y = 500$ .

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

### TOP TIPS

Read the question carefully. Show all your working step by step.  
Check your answer makes sense.

### CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

